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Abstract (Original Investigations)

Title: MULTIMORBIDITY AND TREATMENT OUTCOMES IN ADVANCED CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD): REAL-WORLD EVIDENCE FROM A NATIONAL HEALTH CARE DATASET

Purpose: Chronic obstructive pulmonary disease (COPD) is the 6th leading cause of death in the US. Once diagnosed, medications can slow the progression a prevent acute exacerbations and emergency room visits. Patients with advanced COPD are treated primarily with long-acting beta antagonists (LABA), long-acting muscarinic antagonists (LAMA), and/or Inhaled Corticosteroids (ICS). We aimed to characterize the population with COP receiving dual LAMA+LABA therapy and estimate the effectiveness of changes to treatment regimens. In particular, we evaluated the effect of d escalation to mono-therapy for patients with different COPD severities ar histories of asthma.

Methods: Clinical data of COPD patients treated with LAMA+LABA from 2016-2020 were aggregated from a national insurance claims database, including diagnoses, procedures and prescription fills. Patients with other severe lung diagnoses were excluded (19%). The primary outcome was annualized rate of ER visits for COPD. Regimens were evaluated by estimating the average treatment effect (ATE) using BCAUS, a causal effect machine learning model.

Results: Patients (n=39806) had a mean age of 57.7 (53.0-69.0) and were 59% female. There was a high prevalence of hypertension (68.5%), asthma (34.9%), diabetes (30.9%), and other comorbidities. Smoking and obesity were noted in 62.8% and 32.0% of the cases. Annualized ER visits for COPD were (mean (IQR)): 0.9 (0.0-0.8), ER visits for any diagnosis: 1.4 (0.0-1.5), and COPD exacerbations 0.7 (0.0-1.0). In addition to LAMA+LABA, patients were treated by ICS (67.5%) and Leukotriene Receptor Antagonists (20.5%). Consistent with clinical trial evidence we found that in de-escalation of treatment to monotherapy, LAMA was more effective than LABA in preventing ER visits: -0.70 (-1.60,0.19) but with high variability within and across patient cohorts: 0.39 (-0.19,0.91) for patients with asthma and no exacerbations in the past 12 months vs. -1.64(-3.17,-0.25) with 1 or more exacerbations. For patients without asthma, the comparable values are -0.23(-0.63,0.18) and -1.40(-3.80,1.31), respectively.

Conclusions: Patients treated for COPD have high multimorbidity and medical needs. Estimated treatment effect was highly variable, underscoring the need for the personalized precision medicine paradigm. Findings might be affected by difficulty controlling for confounding variables, heterogeneous effects of specific drugs within a cohort, and study of insured population only.

Feedback and Support

Clinical Implications: Patients treated for COPD have high multimorbidity and medical needs. In a quasi-causal analysis of de-escalation of treatment from LAMA-LABA to monotherapy, LAMA was found to be more effective than LABA in preventing ER visits for COPD.

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Feedback and Support